A Period Formula for Rational Cut-and-Project Strip Projections with Multiplicity

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Abstract

Consider the one-dimensional cut-and-project construction with coprime positive integers α, β defining the rational slope α/β, and window parameter $\omega \geq 0$. Projecting lattice points from the resulting strip onto the line and counting them with multiplicity gives a multiset whose consecutive gaps form a periodic sequence. We prove that its minimal period is

$$\lambda = \begin{cases} N, & D \nmid N, \\ N/D, & D \mid N, \end{cases}$$

where

$$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D = \alpha^2 + \beta^2.$$

The proof reduces the geometry to the distribution of N consecutive residue classes modulo D. When this distribution is non-uniform, the residue classes with larger multiplicity form a proper cyclic interval in the natural residue parametrisation; its only translational symmetry is the identity, which rules out any smaller period. We also discuss how the formula simplifies when multiplicities are discarded. The algebraic core of the proof has been machine-checked in Lean 4.

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1 Introduction

Cut-and-project constructions occupy a central place in the mathematical theory of aperiodic order. The general scheme selects lattice points that fall within a strip (the *window*) around a subspace and projects them onto that subspace, producing point sets whose structure lies between perfect periodicity and complete disorder. For irrational slopes the resulting sequences are non-periodic yet highly ordered (quasicrystals); for rational slopes they are periodic, but the precise period length has received surprisingly little attention in the literature.

The study of these structures was advanced by Yves Meyer's foundational work on model sets and harmonious sets [8], for which he was awarded the Abel Prize in 2017. In his laudatio, Terence Tao presented an illustration of a two-dimensional cut-and-project set and remarked on the distances between the projected points: "They repeat themselves, but not in a regular fashion" [10].

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The present paper determines the period length λ exactly for all rational slopes and all nonnegative real window widths. Section 2 gives the precise construction. Section 3 states the main result. Section 4 contains the proof for the generic case $D \nmid N$. Section 5 treats the degenerate case $D \mid N$. Section 6 derives corollaries. Section 7 describes the machine-checked formalisation in Lean 4. Section 8 surveys related results. Section 9 discusses open problems.

2 Construction of the Sequences

2.1 The cut-and-project set

Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and let $\omega \in \mathbb{R}_{\geq 0}$. Consider the line $f(x) = \frac{\alpha}{\beta}\, x$ in $\mathbb{R}^2$ together with the strip bounded by

$$l(x) = \frac{\alpha}{\beta}(x - \omega) \qquad \text{and} \qquad u(x) = \frac{\alpha}{\beta}\, x + \omega.$$

The set of *accepted lattice points* is

$$\mathbf{C} \;=\; \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^2 \;\middle|\; y \in \left[\frac{\alpha}{\beta}(x - \omega),\; \frac{\alpha}{\beta}\, x + \omega \right] \right\}. \tag{1}$$

2.2 Projection and ordering

The orthogonal projection of $(x, y) \in \mathbb{R}^2$ onto the line f is

$$\mathbf{p} \;=\; \frac{1}{\alpha^2 + \beta^2} \begin{pmatrix} \beta^2 & \alpha\beta \\ \alpha\beta & \alpha^2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{2}$$

The x-coordinate of the projected point is $p_x = \frac{\beta(\beta x + \alpha y)}{\alpha^2 + \beta^2}$. Since the factor $\frac{\beta}{\alpha^2+\beta^2}$ is a positive constant, the ordering of projected points is determined by

$$\tilde{p} \;:=\; \beta x + \alpha y. \tag{3}$$

2.3 Distance sequence and period

Sort the multiset of all $\tilde{p}$-values in ascending order to obtain $(\tilde{p}_s^{(i)})_{i \in \mathbb{Z}}$, and define the *difference sequence*

$$\delta^{(i)} \;:=\; \tilde{p}_s^{(i+1)} - \tilde{p}_s^{(i)}. \tag{4}$$

Since any injective monotone rescaling preserves the period of a sequence, the period length λ of $(\delta^{(i)})$ equals the period length of the Euclidean distance sequence $\big(\|\mathbf{p}_s^{(i+1)} - \mathbf{p}_s^{(i)}\|\big)$.

Definition 2.1. The sequence $(\delta^{(i)})_{i \in \mathbb{Z}}$ has *period length* $\lambda \in \mathbb{N}$ if λ is the smallest positive integer such that $\delta^{(i+\lambda)} = \delta^{(i)}$ for all i.

2.4 The trivial case $\omega = 0$

When $\omega = 0$, the only accepted lattice points lie on f itself: $(x, y) = k(\beta, \alpha)$ for $k \in \mathbb{Z}$. Then $\tilde{p} = \beta \cdot k\beta + \alpha \cdot k\alpha = k(\alpha^2 + \beta^2)$, so all differences equal $\alpha^2 + \beta^2$ and $\lambda = 1$. Note that the formula (6) agrees: $N = 0 + 0 + 1 = 1$ and $D = \alpha^2 + \beta^2 \geq 2$, so $D \nmid 1$ and $\lambda = N = 1$.

3 Main Result

Theorem 3.1 (Period length formula)**.** *Let* $\alpha, \beta \in \mathbb{N}$ *with* $\gcd(\alpha, \beta) = 1$ *and let* $\omega \in \mathbb{R}_{\geq 0}$*. Define*

$$N := \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D := \alpha^2 + \beta^2. \tag{5}$$

Then the period length of the difference sequence (4) *is*

$$\lambda_{\alpha,\beta} = \begin{cases} N & \text{if } D \nmid N, \\ N/D & \text{if } D \mid N. \end{cases} \tag{6}$$

Remark 3.2 (Interpretation of N)*.* The quantity N equals the number of integers in the interval $[-\beta\omega,\ \alpha\omega]$, which is the range of the invariant $r = \alpha x - \beta y$ (see Section 4.1). Geometrically, N counts the number of distinct "tracks" — parallel arithmetic progressions of projected values.

Remark 3.3 (Interpretation of D)*.* The quantity $D = \alpha^2 + \beta^2$ is the common difference of each arithmetic progression of $\tilde{p}$-values. It also equals $\|(\beta, \alpha)\|^2$, the squared length of the direction vector of f.

Remark 3.4 (Computational verification)*.* The formula has been verified against:

- 51,012 test cases from [4] with α, β up to $\sim$100,000 and rational ω up to $\sim$10, all satisfying $D \nmid N$ (51,011 with $N < D$, one with $N \geq D$);
- 468 deliberately constructed cases in the degenerate regime $D \mid N$ (with $\alpha, \beta \leq 10$ and N up to $\sim$500), verifying the $\lambda = N/D$ branch.

All 51,480 cases produce exact matches with zero failures.

4 Proof: The Generic Case $D \nmid N$

4.1 Step 1: Range of the invariant

The strip condition (1), multiplied by $\beta > 0$, reads

$$\alpha(x - \omega) \leq \beta y \leq \alpha x + \beta\omega,$$

equivalently

$$-\beta\omega \leq \underbrace{\alpha x - \beta y}_{=:\, r} \leq \alpha\omega. \tag{7}$$

Since $r \in \mathbb{Z}$, it takes values in

$$\mathcal{R} := \{-\lfloor \beta\omega \rfloor,\ -\lfloor \beta\omega \rfloor + 1,\ \ldots,\ \lfloor \alpha\omega \rfloor\}, \tag{8}$$

using the identity $\lceil -x \rceil = -\lfloor x \rfloor$. Hence $|\mathcal{R}| = \lfloor \alpha\omega \rfloor + \lfloor \beta\omega \rfloor + 1 = N$.

The invariant r is *constant along each track*: if (x, y) and $(x + \beta, y + \alpha)$ are both accepted, they share the same r, since $\alpha(x + \beta) - \beta(y + \alpha) = \alpha x - \beta y$. The step (β, α) is the direction of f.

4.2 Step 2: Each value of r yields one arithmetic progression

For a fixed $r \in \mathcal{R}$, the system

$$\beta x + \alpha y = \tilde{p}, \qquad \alpha x - \beta y = r \tag{9}$$

has determinant $-(\alpha^2+\beta^2) = -D \neq 0$, giving the unique solution

$$x = \frac{\beta\tilde{p} + \alpha r}{D}, \qquad y = \frac{\alpha\tilde{p} - \beta r}{D}. \tag{10}$$

Integrality requires $D \mid (\beta\tilde{p} + \alpha r)$, i.e.

$$\tilde{p} \;\equiv\; c_r \;:=\; -\alpha\beta^{-1} r \pmod{D}, \tag{11}$$

where β^{-1} is the modular inverse of β modulo D (existence: Corollary 4.2). The set of $\tilde{p}$-values for invariant r is the arithmetic progression

$$P_r \;=\; \{c_r + kD : k \in \mathbb{Z}\}. \tag{12}$$

The complete multiset of projected values is $S = \bigcup_{r\in\mathcal{R}} P_r$.

4.3 Step 3: Coprimality lemmas

Lemma 4.1. $\gcd(\alpha, D) = \gcd(\beta, D) = 1$.

Proof. $D = \alpha^2 + \beta^2 \equiv \beta^2 \pmod{\alpha}$, so any common divisor of α and D divides β^2. But $\gcd(\alpha,\beta) = 1$ implies $\gcd(\alpha, \beta^2) = 1$. The argument for β is symmetric. □

Corollary 4.2. *The modular inverse $\beta^{-1} \pmod{D}$ exists, and the map $r \mapsto c_r = -\alpha\beta^{-1} r \bmod D$ is a bijection on $\mathbb{Z}/D\mathbb{Z}$.*

4.4 Step 4: Structure of residues

The N residues $\{c_r : r \in \mathcal{R}\}$ are the image of N consecutive integers under multiplication by $m := -\alpha\beta^{-1} \bmod D$, with $\gcd(m, D) = 1$. Explicitly:

$$\{c_r\}_{r\in\mathcal{R}} \;=\; \{m \cdot r_0,\ m(r_0+1),\ \ldots,\ m(r_0+N-1)\} \bmod D, \tag{13}$$

where $r_0 = -\lfloor \beta\omega \rfloor$.

Lemma 4.3 (Non-uniform residue distribution)**.** *If $D \nmid N$, the multiset of N residues $\{c_r \bmod D : r \in \mathcal{R}\}$ is not uniformly distributed over $\mathbb{Z}/D\mathbb{Z}$. More precisely, writing $N = qD + s$ with $0 < s < D$, exactly s residue classes are hit $q+1$ times and $D - s$ residue classes are hit q times.*

Proof. The map $r \mapsto mr \bmod D$ is a bijection on $\mathbb{Z}/D\mathbb{Z}$ (Corollary 4.2). As r runs through N consecutive integers, the residues $mr \bmod D$ cycle through all D classes with period D. After q complete cycles (qD values), all classes have been hit exactly q times. The remaining $s = N - qD$ values ($0 < s < D$ since $D \nmid N$) hit s additional classes exactly once each. Hence s classes have multiplicity $q+1$ and $D - s$ classes have multiplicity q. □

4.5 Step 5: Period equals N when $D \nmid N$

Within each interval $[kD,\,(k+1)D)$ on the $\tilde{p}$-axis, exactly N points from the multiset S fall (one from each progression P_r). Their positions within the interval are determined by the residues $c_r \bmod D$.

The sorted difference sequence therefore has a repeating block of length N: within each "super-period" of N consecutive elements, the pattern of gaps (including possible zeros from coinciding residues when $N > D$) is identical. Hence $\lambda \mid N$.

Minimality. We now prove that no proper divisor of N is a period, using a *multiplicity function* argument on $\mathbb{Z}/D\mathbb{Z}$.

Define the multiplicity function $\mu\colon \mathbb{Z}/D\mathbb{Z} \to \mathbb{N}_0$ by

$$\mu(c) \ := \ \big|\{r \in \mathcal{R} : c_r \equiv c \pmod{D}\}\big|.$$

By Lemma 4.3, with $N = qD + s$ and $0 < s < D$, the function μ takes exactly two values: $q + 1$ on a set of s residue classes and q on the remaining $D - s$ classes.

It is convenient to pass to the m-stepping parametrisation

$$f(x) := \mu(mx), \qquad x \in \mathbb{Z}/D\mathbb{Z},$$

where $m = -\alpha\beta^{-1} \pmod{D}$. Since m is invertible modulo D, this is just a relabelling of the same data. In this parametrisation, the "heavy" set

$$H := \{x \in \mathbb{Z}/D\mathbb{Z} : f(x) = q + 1\}$$

is a contiguous cyclic interval of length s:

$$H = \{x_0, x_0 + 1, \ldots, x_0 + s - 1\} \pmod{D}$$

for some x_0, because the final s elements of the consecutive block $r_0, r_0 + 1, \ldots, r_0 + N - 1$ contribute one extra hit exactly on s consecutive values in the m-stepping order.

Now suppose, for contradiction, that $\lambda' < N$ is a period of the gap sequence, and write $N = k\lambda'$ with $k > 1$. Let

$$\sigma := \delta^{(0)} + \delta^{(1)} + \cdots + \delta^{(\lambda'-1)}.$$

Since the gap sequence has period λ', we have $\tilde{p}_s^{(i+\lambda')} = \tilde{p}_s^{(i)} + \sigma$ for all i, so the entire multiset S is invariant under translation by σ. On the other hand, N consecutive gaps always sum to D, because every interval $[t, t + D)$ contains exactly N points of S counted with multiplicity. Hence

$$k\sigma = D,$$

so $\sigma = D/k$ is a positive integer and $k \mid D$.

The translation invariance of S implies invariance of the residue multiplicity function:

$$\mu(c + \sigma) = \mu(c) \qquad \text{for all } c \in \mathbb{Z}/D\mathbb{Z}.$$

Equivalently, the relabelled function f satisfies

$$f(x + \tau) = f(x), \qquad \tau := m^{-1}\sigma \not\equiv 0 \pmod{D}.$$

Therefore its heavy set H must satisfy $H + \tau = H$.

But a proper nonempty cyclic interval in $\mathbb{Z}/D\mathbb{Z}$ has trivial stabiliser: if $0 < s < D$ and

$$H = \{x_0, x_0 + 1, \ldots, x_0 + s - 1\} \pmod{D},$$

then $H + \tau = H$ forces $\tau \equiv 0 \pmod{D}$. Indeed, for $1 \leq \tau \leq D - 1$, the translate $H + \tau$ has a different left endpoint and cannot coincide with H unless H is either empty or all of $\mathbb{Z}/D\mathbb{Z}$, neither of which occurs because $0 < s < D$.

This contradiction shows that no proper divisor $\lambda' < N$ can be a period. Since we already know $\lambda \mid N$, it follows that $\lambda = N$. $\square$

5 The Degenerate Case $D \mid N$

Theorem 5.1. *If $D \mid N$, then $\lambda = N/D$.*

Proof. When $D \mid N$, the N values of r span exactly N/D complete cycles of the map $r \mapsto mr \bmod D$. Since this map is a bijection on $\mathbb{Z}/D\mathbb{Z}$, each residue class $c \in \{0, 1, \ldots, D-1\}$ is hit exactly N/D times. Consequently, each progression P_c contributes N/D points per interval of length D on the $\tilde{p}$-axis, and by the uniform distribution, *every* integer $\tilde{p}$ is achieved with multiplicity N/D.

The sorted sequence of $\tilde{p}$-values (with multiplicity) thus consists of each integer repeated N/D times:

$$\ldots,\ \underbrace{t,\ t,\ \ldots,\ t}_{N/D},\ \underbrace{t+1,\ t+1,\ \ldots,\ t+1}_{N/D},\ \ldots$$

The difference sequence is

$$\underbrace{0,\ 0,\ \ldots,\ 0}_{N/D-1},\ 1,\ \underbrace{0,\ 0,\ \ldots,\ 0}_{N/D-1},\ 1,\ \ldots$$

which has period N/D. □

Example 5.2. For $\alpha = 2$, $\beta = 1$, $\omega = 3$: $N = \lfloor 6 \rfloor + \lfloor 3 \rfloor + 1 = 10$ and $D = 5$. Since $5 \mid 10$, the formula gives $\lambda = 10/5 = 2$. Direct computation confirms the difference sequence $(0, 1, 0, 1, \ldots)$ with period 2.

Example 5.3. For $\alpha = 3$, $\beta = 2$, $\omega = 5$: $N = \lfloor 15 \rfloor + \lfloor 10 \rfloor + 1 = 26$ and $D = 13$. Since $13 \mid 26$, the formula gives $\lambda = 26/13 = 2$.

Remark 5.4. The boundary case $N = D$ (i.e. $N/D = 1$) gives $\lambda = 1$: every integer is a projected value with multiplicity 1, and all gaps equal 1.

6 Corollaries

Define $\Lambda_{\alpha,\beta} := \alpha + \beta + 1$.

Corollary 6.1 (Finiteness)**.** *For all $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and all $\omega \in \mathbb{R}_{\geq 0}$, the period length λ is finite.*

Proof. N is a finite positive integer, and $\lambda \leq N$. □

Corollary 6.2 (Period at $\omega = 1$)**.** *For $\omega = 1$, $\lambda = \alpha + \beta + 1 = \Lambda_{\alpha,\beta}$.*

Proof. $N = \alpha + \beta + 1$ and $D = \alpha^2 + \beta^2$. For $(\alpha, \beta) \neq (1, 1)$: $N = \alpha + \beta + 1 < \alpha^2 + \beta^2 = D$ (since $\alpha^2 + \beta^2 - \alpha - \beta - 1 = \alpha(\alpha - 1) + \beta(\beta - 1) - 1 \geq 0$ for $\alpha + \beta \geq 3$), so $D \nmid N$ and $\lambda = N = \alpha + \beta + 1$.

For $(\alpha, \beta) = (1, 1)$: $N = 3$, $D = 2$, $D \nmid N$, so $\lambda = 3 = \Lambda_{1,1}$. □

Corollary 6.3 ($0 < \omega < 1$)**.** *If $0 < \omega < 1$, then $\lambda < \Lambda_{\alpha,\beta}$.*

Proof. $\lfloor \omega\alpha \rfloor \leq \alpha - 1$ and $\lfloor \omega\beta \rfloor \leq \beta - 1$, so $N \leq \alpha + \beta - 1 < \Lambda_{\alpha,\beta}$. Since $\alpha + \beta - 1 < \alpha^2 + \beta^2 = D$ for all $\alpha, \beta \geq 1$ (because $\alpha(\alpha - 1) + \beta(\beta - 1) \geq 0$), we have $N < D$, hence $D \nmid N$ and $\lambda = N < \Lambda_{\alpha,\beta}$. □

Corollary 6.4 ($1 < \omega < 2$)**.** *If $1 < \omega < 2$ and $D \nmid N$, then $\lambda \geq \Lambda_{\alpha,\beta}$.*

Proof. $\lfloor \omega\alpha \rfloor \geq \alpha$ and $\lfloor \omega\beta \rfloor \geq \beta$, so $N \geq \alpha + \beta + 1 = \Lambda_{\alpha,\beta}$. Since $D \nmid N$, we have $\lambda = N \geq \Lambda_{\alpha,\beta}$. □

Remark 6.5. The condition $D \nmid N$ in Corollary 6.4 cannot be dropped. For $\alpha = 2$, $\beta = 1$, $\omega = 3/2$: $N = \lfloor 3 \rfloor + \lfloor 3/2 \rfloor + 1 = 3 + 1 + 1 = 5 = D$. Since $D \mid N$, we get $\lambda = N/D = 1$, which is less than $\Lambda_{2,1} = 4$.

Corollary 6.6 ($\omega > 2$)**.** *If $\omega > 2$ and $D \nmid N$, then $\lambda > \Lambda_{\alpha,\beta}$.*

Proof. $\lfloor \omega\alpha \rfloor \geq 2\alpha$ and $\lfloor \omega\beta \rfloor \geq 2\beta$, so $N \geq 2\alpha + 2\beta + 1 > \Lambda_{\alpha,\beta}$. Since $D \nmid N$, $\lambda = N > \Lambda_{\alpha,\beta}$. □

Remark 6.7. In both corollaries above, the degenerate case $D \mid N$ can produce arbitrarily small periods: $\lambda = N/D$ may be much less than $\Lambda_{\alpha,\beta}$. This occurs when the window width is tuned so that the number of tracks is an exact multiple of $D = \alpha^2 + \beta^2$.

7 Machine-Checked Formalisation

The algebraic core of the proof has been fully formalised in Lean 4 [7] using the Mathlib library [5]. The formalisation comprises approximately 1,300 lines of verified code with no `sorry` axioms, and is available in the companion repository [4] under `Lean/CutAndProject/`.

7.1 Architecture

The proof is structured in two layers. The first layer proves Theorem 3.1 from four abstract axioms collected in a typeclass `GeometricProjection`:

1. $N > 0$ (`N_pos`)
2. The difference sequence has period N (`period_N`)
3. If $D \mid N$, the minimal period is N/D (`period_degenerate`)
4. Every period L induces a residue-preserving translation by some σ with $\sigma N = LD$ (`sigma_of_period`)

Given these axioms, the generic minimality argument (`generic_minimality`, corresponding to Section 4.5) and the case dispatch (`main_theorem`, corresponding to Theorem 3.1) are proved purely algebraically.

The second layer constructs a concrete instance of this typeclass by defining the sorted multiset and difference sequence explicitly. Each axiom is discharged by a separate lemma (`N_pos_concrete`, `period_N_concrete`, `period_degenerate_concrete`, `sigma_of_period_concrete`).

7.2 Correspondence with the paper

Table 1 gives the correspondence between the main results of this paper and the Lean 4 declarations in `Basic.lean`.

7.3 Scope of the formalisation

The formalised proof covers the full algebraic/combinatorial argument: coprimality, residue distribution, cyclic interval structure, the trivial stabiliser lemma, the generic minimality argument, the degenerate case, and the final case dispatch.

The geometric construction (defining the cut-and-project set, projecting lattice points, and sorting the multiset of $\tilde{p}$-values) is modelled abstractly via the `GeometricProjection` typeclass rather than formalised from first principles. Concretely, the sorted multiset is defined combinatorially as $\tilde{p}_s^{(i)} := V(i \bmod N) + \lfloor i/N \rfloor \cdot D$, where V is the quantile function of the cumulative residue distribution. The bridge lemma `count_hits_eq_sorted_count` verifies that this construction is consistent with the residue counting function.

Paper	Lean 4 name	Line
Lemma 4.1 ($\gcd(\alpha, D) = 1$)	`coprime_alpha_D`	11
Lemma 4.1 ($\gcd(\beta, D) = 1$)	`coprime_beta_D`	20
Corollary 4.2 (residue bijection)	`residue_bijection`	43
Lemma 4.3 (non-uniform distribution)	`non_uniform_residue_distribution`	140
Uniform distribution ($D \mid N$)	`uniform_residue_distribution`	215
Heavy set is cyclic interval	`heavy_set_is_cyclic_interval`	269
Trivial stabiliser	`cyclic_interval_stabilizer_trivial`	334
Section 4.5 (generic minimality)	`generic_minimality`	388
Theorem 5.1 ($D \mid N$ case)	`period_degenerate_concrete`	1233
$\sigma N = LD$ from periodicity	`sigma_of_period_concrete`	1206
Theorem 3.1 (main result)	`main_theorem`	1283

Table 1: Correspondence between paper results and Lean 4 declarations.

8 Related Literature

8.1 The three-distance theorem

The three-distance theorem (Steinhaus, Sós, Surányi, Świerczkowski; 1950s) states that placing n points $\{k\theta\}$ on a unit circle produces at most three distinct gap lengths [11, 2]. More recent treatments include the lattice proof by Marklof and Strömbergsson [6].

This theorem is related in spirit but differs from Theorem 3.1 in two essential ways:

- It concerns irrational rotations on a circle, not lattice projections through a strip of finite width.
- It characterises the *number of distinct gap lengths* (at most three), not the *period length* of the gap sequence.

8.2 Christoffel words and Sturmian sequences

The lower Christoffel word of slope p/q with $\gcd(p, q) = 1$ is a binary word of length $p + q$, and the mechanical (Sturmian) sequence of rational slope p/q has period $p + q$ [1].

The difference from Theorem 3.1 is significant:

1. Christoffel words have period $p + q$; at $\omega = 1$ our result gives $\alpha + \beta + 1$. The extra $+1$ arises from the symmetric window of unit width.
2. The Sturmian framework describes a single line (vanishing window width), not a strip of finite width.
3. The present result concerns Euclidean distances between projected points, not a symbolic word over a finite alphabet.

8.3 Cut-and-project sets and model sets

The mainstream literature on model sets [9, 8, 3] focuses almost exclusively on irrational slopes, since rational slopes yield periodic sequences considered less interesting from the perspective of aperiodic order. No prior work was found that determines the period of the distance sequence for a one-dimensional cut-and-project set with rational slope and a symmetric window of finite width.

9 Discussion

9.1 Why the formula splits into two cases

The key structural distinction is whether the N tracks (arithmetic progressions) tile the residues modulo D uniformly or not:

- When $D \nmid N$: the tracks distribute non-uniformly among the D residue classes. The resulting interleaving pattern is genuinely aperiodic at any scale smaller than N, giving $\lambda = N$.
- When $D \mid N$: each residue class is hit exactly N/D times. Every integer is a projected value (with multiplicity N/D), collapsing the gap structure to a simple pattern of period N/D.

9.2 Open problems

1. **Dependence on the window position.** The present paper considers a strip in the specific position used in (1). If one translates the window transversally, the set of admissible invariant values changes by a shifted interval, but boundary effects may alter the interleaving pattern. It would be interesting to determine whether the same period formula persists for arbitrary window offsets.
2. **Higher dimensions.** The cut-and-project construction generalises to higher-dimensional lattices. Whether analogous period formulas hold in dimension $d \geq 2$ is unexplored.
3. **Distribution of gap lengths.** For fixed α, β, ω, one may ask for a precise characterisation of the distinct gap values and their multiplicities within one period.

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